MIL-A-15373B(SHIPS)
26 June 1964
SUPERSEDING
MIL-A-15373A(SHIPS)
31 March 1958
(See 6.4)

MILITARY SPECIFICATION

ANNUNCIATORS, NAVAL SHIPBOARD

1. SCOPE

1.1 Scope.- This specification covers annunciators for use aboard ship with the officers call bell system and with the sound-powered telephone and voice tube call bell system.

1.2 Classification.- The annunciators shall be of types designated in the following form, as specified (see 6.1):

IC/A4B

Where IC/ is the basic indicator; the letter symbol "A" indicates the supply voltage and frequency (see 1.2.2); the digit symbol "4" indicates the number of indicators (see 1.2.3); and the letter symbol "B" indicates the number of relays (see 1.2.4).

1.2.1 Basic indicator.- The basic indicator for Naval interior communication equipment shall be "IC/".

1.2.2 Supply voltage and frequency.- The supply voltage and frequency rating of annunciators shall be identified as follows:

Symbol	Voltage and frequency
A	115 volts, d.c.
B	115 volts, a.c., 60 cycles
C	115 volts, a.c., 400 cycles
D	23 volts, 1000 cycles[1/] (d.c. lighting and reset)
E	23 volts, 1000 cycles[1/] (a.c. 60 cycles lighting and reset)
F	23 volts, 1000 cycles[1/] (a.c. 400 cycles lighting and reset)

1/ Power to be supplied from a hand-driven call signal station similar to that shown on Drawing 9000-S6501-73801.

1.2.3 Indicators.- The number of indicators shall be identified as follows:

Symbol	Number of indicators
4	4
9	9
16	16
25	25

1.2.4 Relays.- The number of relays shall be identified as follows:

Symbol	Number of relays
A	0
B	1
C	2
D	3

FSC 6350

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-983 - Interior Communication Equipment, Naval Shipboard; Basic Design Requirements for.
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment, Naval Shipboard Use.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

BUREAU OF SHIPS

9000-S6501-73801 - Call Signal Station, Type IC-D.
9000-S6504-73992 - Indicator and Relay for Type IC/Annunciators.
9000-S6504-73993 - Annunciator Enclosure, 4-Indicators.
9000-S6504-73994 - Annunciator Enclosure, 9-Indicators.
9000-S6504-73995 - Annunciator Enclosure, 16-Indicators.
9000-S6504-73996 - Annunciator Enclosure, 25-Indicators.
S6504-1,449,019 - Wiring Diagrams for Type IC/Annunciators.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample.- Prior to beginning production a sample of each type of annunciator shall be tested as specified in 4.3 (see 6.3).

3.2 General features.- The equipment shall be in accordance with the following paragraphs of MIL-I-983, in addition to the requirements specified herein. (Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.)

Materials, general
Substitution of (equal or superior) materials or parts
Fungus-inert materials
Unacceptable materials
Flammable materials
Toxic materials
Wood
Aluminum
Magnesium
Iron and steel
Nonferrous material (except aluminum)
Plastics
Lubricants and lubrication
Painting
Protection against corrosion
Bolts, machine screws, studs and nuts
Gaskets
Locking devices
Washers
Parts - electrical - general

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Variable resistors
Relays
Electrical tapes
Batteries
Dial illumination lamps
Enclosures - general
Enclosure - accessibility
Enclosure - degree of (watertight - 3-feet)
Minimum sheet metal thicknesses (watertight enclosures)
Through bolting
Cable entrance
Threaded devices
Rounded corners and edges
Internal subassembly protection
Shock, vibration and inclination (nonvital vibration test required)
Drilled and tapped holes
Welding
Temperature and humidity (special cold limit applies)
Salt spray
Primary power supply circuits
Power supply tolerances
Personnel protection
Ground potential and grounding
Soldering
Electrical parts mounting
Terminal boards, connectors and terminals
Wiring
Dielectric strength and insulation resistance clearances
Interchangeability and standardization
Designation and marking
Workmanship and general examination

3.3 Design.- Annunciators shall be in accordance with this specification and as shown on Drawings 9000-S6504-73992 to 9000-S6504-73996, inclusive, and S6504-1,449,019, as applicable.

3.4 Contact resistance.- The contact resistance of the indicators and relays shall not exceed 0.015 ohm when measured in accordance with 4.6.1.

3.5 Endurance.- The annunciator shall be capable of withstanding, without mechanical or electrical damage, an endurance test, including the following features:

Power supply voltage and frequency variation (see MIL-I-983).
Endurance (24,000 cycles).
Temperature and humidity cycles (see MIL-I-983).

The details of this test shall be as specified in 4.6.2.

3.6 Sound output.- The audible signal of the 23-volt, 1000-cycle annunciators (see 1.2.2) shall have a fundamental frequency of between 750 and 1000 cycles per second. The audible signal shall have a minimum sound output of 95 db relative to 0.0002 dyne per square centimeter when tested in accordance with 4.6.4.

3.7 Repair parts.- Onboard repair parts shall be determined by the type and quantity of annunciators being furnished per ship. Repair parts shall include one of each type of the following parts for each ten or fraction thereof installed:

Indicator assembly
Relay assembly
Transformers
Rectifiers
Rheostats
Resistors
Capacitors
Lamps

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The requirements for the determination, identification, and procurement of stock repair parts and for revision of the quantities of onboard repair parts shall be as specified in MIL-P-15137.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General inspection.- The procedures for the various tests to be performed on the equipment shall be as specified hereinafter.

4.3 Preproduction inspection.- Preproduction inspection shall consist of the examination and tests specified in table I, which shall be performed, in general, in the order listed. One unit of each type or a composite nine indicator annunciator consisting of the following: three indicators and associated relay to operate on 115 volts, a.c., 60 cycles; two indicators and associated relay to operate on 115 volts, a.c., 400 cycles; one indicator and associated relay to operate on 115 volts, d.c.; and three indicators to operate from the output of a standard Navy call signal station as shown on Drawing 9000-S6501-73801, shall be inspected.

Table I - Preproduction inspection.

Examination and tests	Requirement paragraph	Test reference
General examination	3.2	MIL-I-983
Contact resistance	3.4	4.6.1
Vibration (for nonvital equipment)	3.2	MIL-I-983
Inclination	3.2	MIL-I-983
Temperature and humidity[1/]	3.2	MIL-I-983
Supply line voltage and frequency variation[1/]	3.2	MIL-I-983
Endurance[1/]	3.5	4.6.2
Salt spray	3.2	MIL-I-983
Dielectric strength	3.2	4.6.3
Insulation resistance	3.2	MIL-I-983
Degree of enclosure	3.2	MIL-I-983
Class HI shock	3.2	MIL-I-983
Output sound pressure[2/]	3.6	4.6.4
Contact resistance	3.4	4.6.1

[1/] These tests shall be performed concurrently.
[2/] Required for 23-volt, 1000-cycle supply annunciators only.

4.4 Sampling for quality conformance inspection.-

4.4.1 Lot.- All annunciators of the same type designation offered for delivery at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period.

4.4.2 Sampling for examination and tests.- Sample annunciators shall be selected in accordance with MIL-STD-105 at inspection level II for the inspection specified in 4.5. The Acceptable Quality Level (AQL) shall be 1.5 percent defective.

4.5 Quality conformance inspection.- The sample annunciators selected in accordance with 4.4.2 shall be subjected to the examination and tests specified in table II. Tests shall be performed, in general, in the order listed.

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Table II - Quality conformance inspection.

Inspection	Requirement paragraph	Test reference
General examination	3.2	MIL-I-983
Dielectric strength	3.2	4.6.3
Insulation resistance	3.2	MIL-I-983
Degree of enclosure	3.2	MIL-I-983
Contact resistance	3.4	4.6.1
Output sound pressure	3.6	4.6.4

4.6 Test procedures.-

4.6.1 Contact resistance.- The contact resistance through a stationary contact and moving contact that form part of the same circuit shall be determined by an ohmmeter. Ohmmeter readings shall be taken prior to the endurance tests and upon completion of the endurance test.

4.6.2 Endurance.- The annunciator shall be operated for 24,000 cycles. A cycle shall consist of operating all of the indicators consecutively at a rate of at least one indicator per second, and then resetting all indicators. This test shall be continuous and shall be conducted in accordance with table III.

Table III - Endurance test.

Cycles[1/]	Power supply							Relative humidity
	Nominal 115 volts d. c.	Nominal 115 volts a. c. 60-cycle		Nominal 115 volts a. c. 400-cycle		Nominal 23 volts 1000-cycle Generator crank speed	Temperature	
	Voltage	Voltage	Frequency	Voltage	Frequency	R. p. m.	Degrees F.	
0-5000	115	115	60	115	400	212	75 (room)	Not controlled
5001-9000	115	115	60	115	400	212	-35	Not controlled
9001-14000	115	115	60	115	400	212	131	Approx. 90 percent for alternate groups of 500 cycles each
14001-16000	122	122	65 55	122	415	229	75 (room)	Not controlled
16001-18000	108	108	55	108	385	194	75 (room)	Not controlled
18001-24000	115	115	60	115	400	212	75 (room)	Not controlled

[1/] Cycles to be adjusted to suit laboratory test schedule.

4.6.3 Dielectric strength.- The dielectric strength test shall be conducted in accordance with MIL-I-983, except that an r. m. s. test voltage of 900 shall be used for all tests.

4.6.4 Sound output pressure.- Sound output pressure tests shall be conducted in accordance with MIL-I-983. A standard distance of 10 feet shall be used.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.- The equipment, accessories, technical publications, and repair parts shall be preserved and packaged by level A or C; packed by level A, B, or C and marked by level A or B or C as specified (see 6.1) in accordance with MIL-E-17555.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of annunciators required (see 1.2).
(c) Level of preservation and packaging, packing and marking required (see 5.1).

6.2 Stocked types.- Annunciators will be stocked in the special types indicated in table IV, suitable for one, two or three relays.

Table IV - Stocked types of annunciators.

Type	Voltage and frequency		Number of indicators	Number of relays
	Volts	Cycles		
IC/A4D	115 D.C.		4	Wired for 3 equipped with 1
IC/A9D	115 D.C.		9	Wired for 3 equipped with 1
IC/A16D	115 D.C.		16	Wired for 3 equipped with 1
IC/B4D	115	60	4	Wired for 3 equipped with 1
IC/B9D	115	60	9	Wired for 3 equipped with 1
IC/B16D	115	60	16	Wired for 3 equipped with 1
IC/E4A	23	1000	4	0

6.2.1 Annunciators of any classification may be procured from a manufacturer for a specific end use and may include combinations of indicators and relays having different supply voltages and frequencies.

6.3 Preproduction.- Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - Ships
(Project 6350-N009Sh)

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 69 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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